

TIC 447115398

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	23.062962 d (ALIASED, see below)	VSX	no match
-> corrected period:	46.125925 d	ASAS-SN	no match
Epoch (t0)	2368.5185 BTJD	Gaia eclipsing-binary table	no match
Duration	0.1200 d	Gaia variable classification	no match
Depth	10.830 % (108,302 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	422.2	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	9 (gate >= 3)		
Robust depth ratio	1.04 (gate >= 0.50)		
Robust depth sigma	426.1 (gate >= 7.0)		
Odd/even z-score	11.60 (ALIAS FLAGGED)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

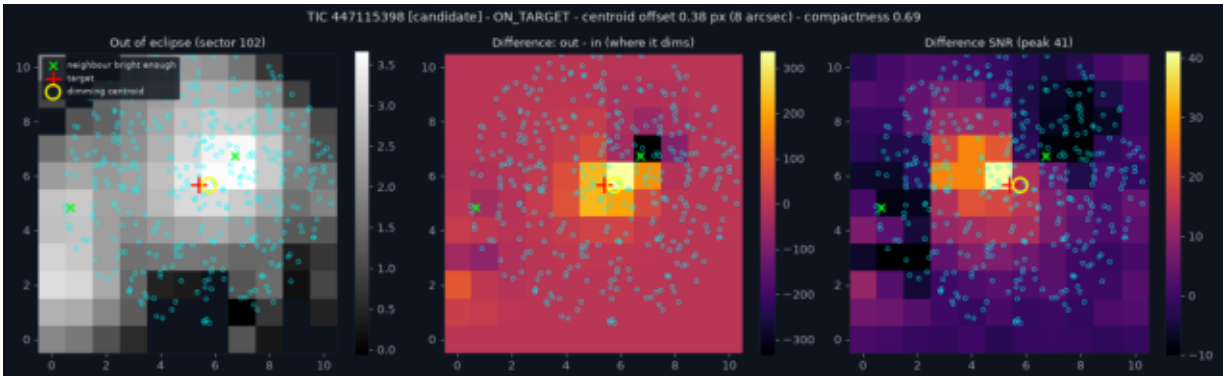
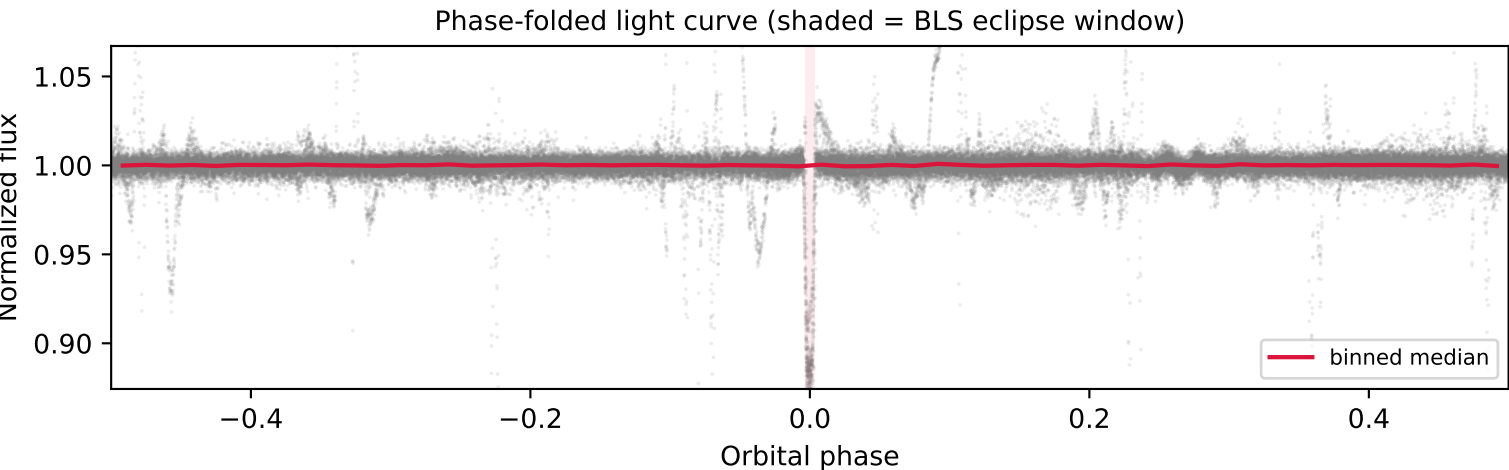
Sector 102 light-curve dip in this sector: 125.9 sigma t0 refined by 0.0000 phase

Difference-image peak SNR 41.18 (gate >= 5.0) Compactness 0.69 (gate >= 0.50)

Dimming centroid offset 0.38 px (8 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 447115435

1.44 px away, dTmag -1.27 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 11.60 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 46.125925 d, in any write-up.